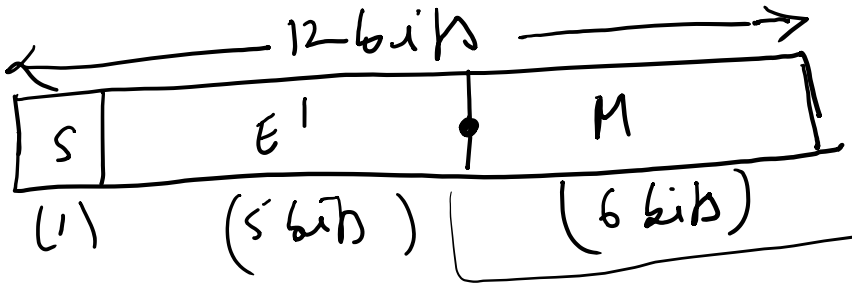
Consider the following 12-bit FP number representation.



Perform sub, Add, Mul & Divide on operands:

A = 0 10001 01101

B = 1 01111 10101 0

1 → +ve

B → -ve

$E' = 5 = 2^5 = \frac{32}{2} = 16 - 1 = 15$

$E' = E' = (8) = 2^8 = \frac{256}{2} = 128 - 1 = 127$

SP

$E' = (11) = 2^{11} = \frac{2^{11}}{2} = 2^{10} = 1024 - 1 = 1023$

DP

$S_A = 0$ $E'_A = 10001 = 17_{10}$ $M_A = 01101$

$S_B = 1$ $E'_B = 01111 = 15_{10}$ $M_B = 10101$

$E_A = 17 - 15 = 2$

$E_B = 15 - 15 = 0$

$Value_A = +1.011011 \times 2^2$

$Value_B = -1.101010 \times 2^0 \xrightarrow{\text{Shift}} -0.01101010 \times 2^2$

I : Sub : $A - B = A - (-B) = A + B$

* Shift M_B right by $|2-0| = 2$ bits

$$
\begin{array}{rl}
A = & +1.011011 \times 2^2 \\
B = & -0.01101010 \times 2^2 \\
\hline
 & +1.10101110 \times 2^2
\end{array}
$$

Truncate (Round)

6 bits

1.101011×2^2

guard bits

$+ \quad 0.000001$

$1.101110 \times 2^2 = E$

$E' = 15 + 2 = 17$

0	10001	101110

$\Rightarrow$ Result

II Add

$S_A = 0 \quad E_A' = 17 \quad E_A = 2 \quad M = 011011$

$S_B = 1 \quad E_B' = 15 \quad E_B = 0 \quad M = 101010$

A − B

$A = +1.011011\overset{\curvearrowright}{1}00 \times 2^2$

$B = -0.0110101\,0 \times 2^2$

$+1.000000\,10 \times 2^2$ → Rounding

$+ \quad 0.000001$

$+1.000001 \times 2^2 \Rightarrow E$

$E' = 15+2$

0	10001	000001

⇒ Result

III Multiply

* Add exponents
* Sub bias
* multiply mantissa
* Normalize / truncate if required

$E_A' = 17$

$E_B' = 15$

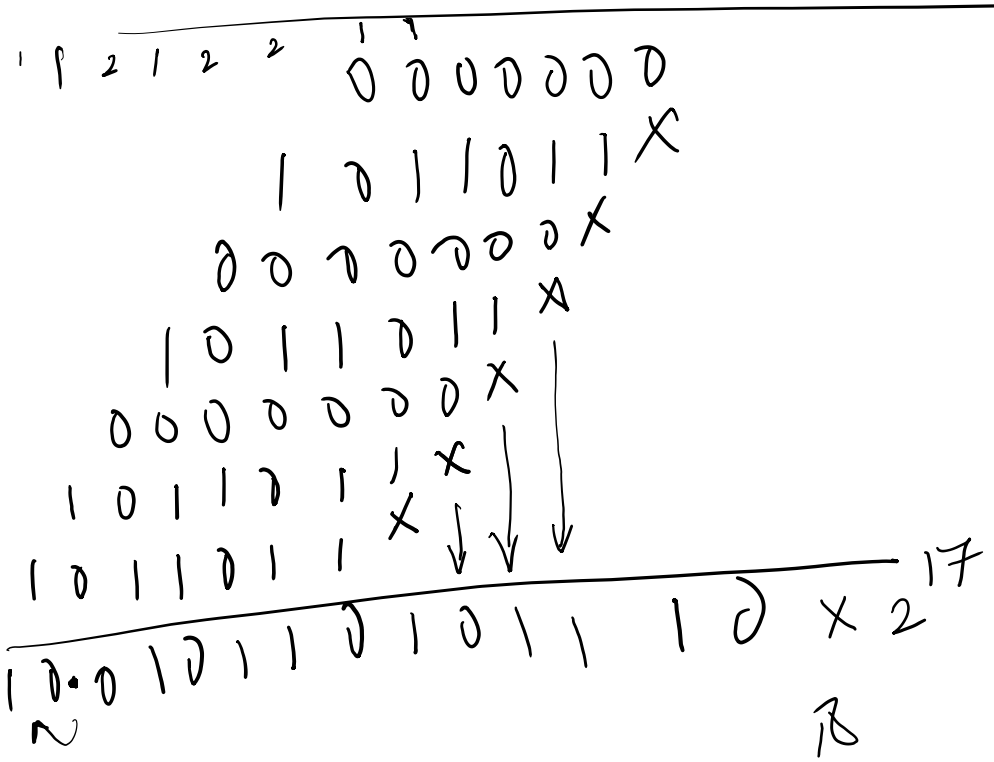
32

$-\ 15$

17

A B

1.011011×1.101010



$\varphi = 100$

$\zeta = 101$

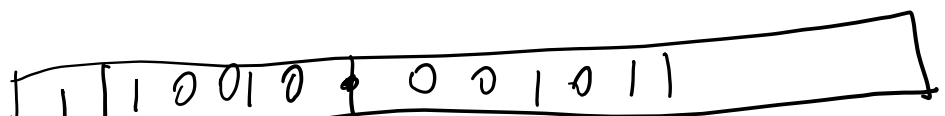
$10.010110101110 \times 2^{17}$

$1.001011010111 \times 2^{18}$

→ guard bits, truncate using Rounding

-1.001011×2^{18}

Res =



S_R E'_R M_R

Divide

* subtract exponents
* add bias
* Divide mantissa
* Normalize & truncate if required

Res = 